Yicheng Shen

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Scientific Interests

Real-world evidence synthesis, comparative effectiveness research, target trial emulation, digital twin and individualized health prediction using observational data, electronic health records, and multi-institutional cohorts.

Methodological Interests

Causal inference, Bayesian hierarchical modeling, multivariate longitudinal modeling, survival analysis, federated and distributed learning, transfer learning, dictionary learning, missing data and (network) meta-analysis.

Education

Johns Hopkins University — Baltimore, MD
Ph.D. in Biostatistics, GPA: 4.00/4.00 — Aug 2024 – May 2029
Advisors: Dr. Jiayi (Jessie) Tong & Dr. Elizabeth A. Stuart

Duke University — Durham, NC
M.S. in Statistical Science, GPA: 3.98/4.00 — Aug 2022 – May 2024
Advisors: Dr. Hwanhee Hong & Dr. Alexander Volfovsky

Carleton College — Northfield, MN
B.A. in Statistics & Political Science, GPA: 3.9/4.0 — Aug 2018 – Jun 2022
Advisors: Dr. Andrew Poppick & Dr. Greg Marfleet
Graduate with Distinctions & *Magna Cum Laude*

Selected Coursework

Statistics: Bayesian statistics, time series analysis, predictive modeling, statistical inference, hierarchical modeling, survival analysis, statistical computing, causal inference, probabilistic machine learning, spatio-temporal modeling, high-dimensional statistics.

Mathematics: multivariate calculus, linear algebra, mathematical structure, ordinary differential equations, numerical analysis, computational mathematics, real analysis, measure theoretical probability.

Working Papers & Publications

Shen, Y., Stuart, E. A., Iwashyna, T. J., Hochberg, C., Tong, J. (2026). Scaling Real-World Evidence in Critical Care: A Federated Clone-Censor-Weight Framework for Target Trial Emulation. [Manuscript in preparation for *Journal of the American Statistical Association: Applications and Case Studies*]

Shen, Y., Kim, J., Luo, C., Zeger, S. L., Shah, A. A., Tong, J. (2026). MV-PEAL: A Federated Learning Framework for Multivariate Longitudinal EHR Data. [Invited Revision at the *Annals of Applied Statistics*]

Shen, Y., Kim, J., Luo, C., Zeger, S., Domsic, R. Shah, A., Tong, J. (2026). Unlocking Multi-Institutional Insights into Disease Progression with Federated Learning: PEAL as a Lossless, One-Shot Solution. *NPJ Digital Medicine*. https://doi.org/10.1038/s41746-026-02861-6

Vazquez, J., **Shen, Y.**, Sanderson, K., Akulian, J., Tong, J., Stuart, E. A.(2026). Federated Learning with Incomplete Data: A Weighted Approach. https://doi.org/10.48550/arXiv.2605.20125

Wu, X., **Shen, Y.**, Shan, M., Stuart, E., Lipkovich, I. (2026). Comparing Federated Learning Methods for Estimating Average Treatment Effects in a Target Population from Multiple Real-world Studies and Randomized Trials. [Submitted for *Statistics in Medicine*]

Chao, A., **Shen, Y.**, Ghanta, A., Silver, N., Milardo, R., Santos, C., Tong, J. (2025). Sex Differences in Response to Incretin-Based Medications for Obesity: Systematic Review and Meta-Analysis. [Invited Revision at *Obesity Reviews*]

Shen, Y., Chu, H., Hong, H. (2025). Simulation-Based Methods for Power Calculation in Bayesian Network Meta-Analysis. [Invited revision at *Journal of the Royal Statistical Society Series C*]

Shen, Y., Bail, C., & Volfovsky, A. (2026). Enhancing joint ideal point estimation strategies with Twitter social networks and text data. *Network Science*, 14, e10. doi:10.1017/nws.2026.10028

Shen, Y. (2022). Be Steady and Popular: a Modern Counter-Insurgency ABM. *NetLogo Modeling Commons.*

Shen, Y. (2022). Justified Cause? Assessing the Humanitarian Outcomes of US Foreign Aid and Intervention Since the Cold War. *PoliS: Carleton Journal of Political Science*, inaugural edition, pp.129-167.

Shen, Y., Flignor, J., Nachreiner, L., & Wang, K. (2022). Behind the Smoke: An Extreme Value Analysis of Air Pollution in Minnesota. *Consortium for the Advancement of Undergraduate Statistics Education*, spring 2022 issue.

Published Software

Luo, C., Duan, R., ⋯, **Shen, Y.** , Tong, J., & Chen, Y. (2025). PDA: Privacy-Preserving Distributed Algorithms. R Package. CRAN Package Site.

Shen, Y. (2022). Model of Wheeler. NetLogo Modeling Commons. Northwestern University Center for Connected Learning and Computer-Based Modeling, Evanston. Netlogo Modeling Commons Site.

Coauthored Chapter

Knoke, D. (2025). Insurging. In *Network Collective Action: Agent-Based Models of Pandemics, Riots, Social Movements, Insurrections and Insurgencies* (pp. 89-100). Cham: Springer Nature Switzerland. Simulation Software.

Honors, Awards & Funding

New England Rare Disease Statistics BEST Paper Award	2026
ICSA Applied Statistics Symposium Student Paper Award	2026
PHAISE AI Research Day Student Award	2026
NSF Travel Award for the ICSA Applied Statistics Symposium	2026

Honorable Mention for the BEST Award for Master's Research	2024
Teaching Assistant of the Year Award	2024
Dean's Research Award for Master's Students (twice)	2022 & 2023
TADA Travel Award	2023
Duke Career Center Professional Development Fund	2022
First Prize in Undergraduate Statistics Research Project Competition	2022
Phi Beta Kappa Society	2022
Sigma Xi Society	2022
Distinction on Statistics Senior Thesis	2022
Distinction on Political Science Senior Thesis	2022
Dean's Honor List - Top 10% of Class	2021

Teaching Experience

Teaching Assistant (Johns Hopkins University) Aug 2025 – Present
140.751-2: Advanced Methods in Biostatistics
140.778: Statistical Computing, Algorithm, and Software Development
140.779: Advanced Statistical Computing
140.641: Survival Analysis
140.622: Statistical Methods in Public Health

Teaching Assistant (Duke University) Aug 2023 – May 2024
STA 610: Multilevel and Hierarchical Models
STA 432: Theory and Methods of Statistical Learning and Inference

Teaching Assistant (Carleton College) Sep 2020 – May 2022
STA 320: Time Series Analysis
STA 230: Applied Regression Analysis
STA 220: Introduction to Data Science

Work Experience

Quantitative Resource Center Northfield, MN
Comprehensive Integrative Exercise Consultant Sep 2021 – June 2022

Undergraduate Journal of Humanistic Studies Northfield, MN
Editor of 2020 & 2021 Issues Mar 2021 – June 2022

Minnesota Council of Nonprofits Saint Paul, MN
Legislator Data Analytic Internship Nov 2020 – Dec 2020

Northfield Area Family YMCA Northfield, MN
Statistical Consultant Jan 2021 – Mar 2021

Laurence McKinley Gould Library Northfield, MN
Special Collection and Preservation Assistant Oct 2018 – Apr 2022

Invited Talks & Conferences

New England Rare Disease Statistics Workshop (NERDS)	2026
Joint Statistical Meeting (JSM)	2026
ICSA Applied Statistics Symposium	2026
Eastern North American Region (ENAR) Spring Meeting	2024, 2026
Johns Hopkins inHealth Precision Medicine Symposium	2026

	PHAISE AI Research Day	2026
	Duke Industry Statistics Symposium (DISS)	2024, 2026
	Duke Research Data Visualization Competition and Showcase	2024
	Duke Health Data Science Poster Showcase	2023
	New Directions in Analyzing Text as Data (TADA)	2023
	Electronic Undergraduate Statistics Research Conference	2022
	Northfield Undergraduate Mathematics Symposium (NUMS)	2021
	Carleton Undergraduate Research and Internship Symposium	2021
Organized Session	ENAR Invited Session: Federated Learning at the Frontier	2027

Membership

International Chinese Statistical Association
New England Statistical Society
Eastern North American Region of International Biometric Society
International Society for Bayesian Analysis
American Statistical Association

Service

Reviewer for *Epidemiologic Methods*
Reviewer for *Statistics in Biopharmaceutical Research*
Departmental Ph.D. Student Representative (Hopkins)
Organizer of Biostatistics Education & Student Training Seminar (Hopkins)
Organizer of Johns Hopkins Causal Inference Working Group (Hopkins)
Undergraduate Student Departmental Advisor (Carleton)

Research / Peer Mentoring

Xiao Wu (MS)
Hui Yao (MS)
Yifan Zhang (PhD)

Skills

Programming
Proficiency in: R, Shiny, Python, MySQL, MATLAB, Mathematica, NetLogo.

Software & Statistical Implementation
Rich experience in GitHub, Quarto, LaTeX & Overleaf, Jupyter Notebook, GWAS with PLINK, coding and operating on server clusters, predictive modeling, hierarchical modeling, agent-based modeling, clustering and classification problems, JAGS, MCMC-related Bayesian sampling and diagnostics.

Personal Interests

Karate, badminton, movies, hiking, volunteer work, rare book preservation.